

Real or Robotic? Assessing Whether LLMs Accurately Simulate Qualities of Human Responses in Dialogue

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Abstract

Studying and building datasets for dialogue tasks is both expensive and time-consuming due to the need to recruit, train, and collect data from study participants. In response, much recent work has sought to use large language models (LLMs) to simulate both human-human and human-LLM interactions, as they have been shown to generate convincingly human-like text in many settings. However, to what extent do LLM-based simulations *actually* reflect human dialogues? In this work, we answer this question by generating a large-scale dataset of 100,000 paired LLM-LLM and human-LLM dialogues from the WildChat dataset and quantifying how well the LLM simulations align with their human counterparts. Overall, we find relatively low alignment between simulations and human interactions, demonstrating a systematic divergence along the multiple textual properties, including style and content. Further, in comparisons of English, Chinese, and Russian dialogues, we find that models perform similarly. Our results suggest that LLMs generally perform better when the human themselves writes in a way that is more similar to the LLM’s own style.

1 Introduction

Large language models (LLMs) are capable of producing convincingly human-like responses to a broad range of inputs. Recent work has explored the potential of LLMs to simulate human interactions in different scenarios (Zheng et al., 2023; Köpf et al., 2024; Zhao et al., 2024). The scenarios include simulating humans interacting with other humans to generate dialogue datasets, as well as simulating humans interacting with other LLMs (Kim et al., 2024), which is thought to be a scalable, automated approach for LLM quality testing. Such simulations can greatly reduce the cost of

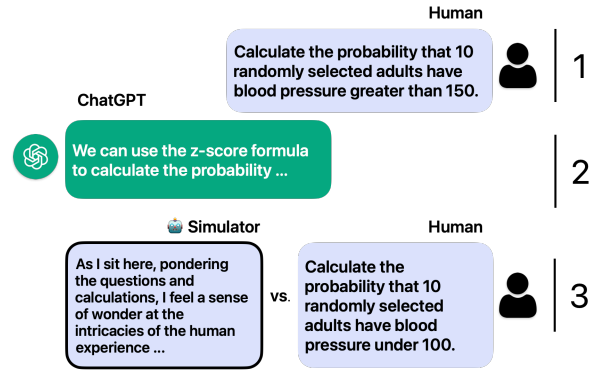


Figure 1: A sample conversation between a human and GPT-3.5 on WildChat and LLAMA3.1-8B’s simulation of Turn 3 of the conversation. In this study, we compare the SIMULATOR’s output against the HUMAN’s output using 21 metrics, covering lexical, syntactic, semantic, and stylistic features.

collecting these data, which often require costly human labor and are difficult to scale to the diversity of LLM abilities. However, this approach can only be effective if the responses generated by the LLM mirror how a human would interact in different scenarios. In this work, we ask: to what extent can LLMs simulate the responses of humans in a human-LLM dialogue?

To test simulation capabilities, we evaluate the degree to which LLMs mirror human behavior in real Human-LLM dialogues. Our study asks the following three research questions: (**RQ1**) to what extent does the choice of model and prompt instructions influence how well the LLM can simulate human behavior; (**RQ2**) how do these results generalize to interactions in languages other than English; and (**RQ3**) in what contexts are LLMs more likely to effectively simulate human responses?

To answer these questions, we develop an evaluation framework on top of human responses from the one million conversations in the WildChat dataset (Zhao et al., 2024). We compare the response from

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a human in a dialogue with the simulated response by an LLM, which we denote as a SIMULATOR, as illustrated in Figure 1. Responses are compared across multiple categories, e.g., lexicality, syntax, semantics, and style, to assess their fidelity to human behavior. Using multiple LLMs and instruction prompts, we evaluate which settings led to better simulation; and, by using regression methods, we highlight the most significant factors that lead SIMULATOR responses to be more human-like.

This paper makes the following four contributions. First, we introduce a general evaluation framework for meaningful analysis of human-LLM simulations and a new dataset of over 1.2K annotator responses to the same conversations, providing a human-level performance comparison. Second, we perform a large-scale analysis of 9 LLMs across 50 prompts simulating 2K English human-LLM conversations, and even the best model and prompt combinations are relatively weak at simulating human behavior. Third, multilingual analyses on 10K Chinese and Russian human-LLM conversations show that performance is roughly similar, though still low. Fourth, a regression analysis demonstrating which factors lead to more human-like responses from LLMs, we find that when the human begins the conversation in a writing style that resembles the LLM’s, the SIMULATOR can better match their behavior in a later turn. All data and code are released for research purposes at <https://github.com/davidjurgens/human-llm-similarity>.

2 Using LLMs to Simulate Human Interaction

Given the practical use of LLMs in mimicking human turns in conversations, many studies deal with simulating human-LLM interactions and developing relevant conversational data sets for generic conversations (Tamoyan et al., 2024; Njifenjou et al., 2024; Gosling et al., 2023) or for conversations of a specific domain, such as education (Abasiantaeb et al., 2024), health (Cho et al., 2023), or programming (Liu et al., 2024b). Similarly, LLM-LLM interactions highlight the general capability of LLMs to mimic human discussions (Park et al., 2023; Zhou et al., 2024; Rossetti et al., 2024; Zhou et al., 2023; Chen et al., 2023). In order to facilitate further research, a battery of studies have introduced various datasets of human-LLM (Zheng et al., 2023) and LLM-LLM (Kim et al., 2022; Chen et al., 2023) dialogue. Another branch of

interaction simulation has studied Digital Twins as virtual replicas of physical systems, in this case, humans in various discussion settings (Barricelli et al., 2019; Barricelli and Fogli, 2024; Rossetti et al., 2024; Wen et al., 2024). It is common in many of these contexts for LLMs to augment—or even replace—human labor with simulations, thereby reducing time and effort (Kojima et al., 2022; de Wit, 2023).

Humans, however, tend to have their specific style, intent, and self-creativity, which are challenging for LLMs to mimic despite recent technological breakthroughs (Stevenson et al., 2022; Wu et al., 2023; Wolf et al., 2023; Gui and Toubia, 2023; Jiang et al., 2024b). Specifically, Leng and Yuan (2023) highlight that while LLMs show promise for applications in social science research, further investigation into their subtle behavioral differences from humans and the development of robust evaluation protocols are essential, thus motivating our research. Multiple metrics have been proposed to measure these differences, including content relevance (Abeyasinghe and Circi, 2024), emotional alignment (Mæhlum et al., 2024), and intent accuracy (Kim et al., 2024). A few studies (Sedoc et al., 2019; Svikhnushina and Pu, 2023) have explored broader evaluation techniques by developing frameworks that align closely with human judgment. However, comprehensive research about how different measures vary between LLMs and prompts is still lacking. In this work, we address this measurement gap by analyzing how LLM responses compare to human responses across multiple similarity measures, LLM models, and prompts.

3 Problem Definition

Our research questions focus on understanding which settings lead to SIMULATORS producing convincingly HUMAN-like dialogues. We adopt the following modeling task, which provides a controlled setting for this question: Given the initial human utterance in a conversation, followed by the response of a chatbot, we prompt a SIMULATOR to suggest the next HUMAN response in the discussion. The “true” value of this third turn is known (but not given as input to the SIMULATOR), as it is part of the WildChat dataset and thus acts as a reference response to which we compare.

This modeling setup offers the following desirable properties. First, by using HUMAN Turn 3 as a reference, we are able to investigate multiple fac-

tors related to the quality of LLM simulations: (i) the similarity between SIMULATOR and HUMAN responses along a variety of linguistic features, (ii) the influence of the original human Turn 1 and chatbot Turn 2 utterances, and (iii) the influence of model and prompt. Second, limiting to HUMAN Turn 3 allows us to study these factors *in isolation at the early stage of dialogue*, i.e., without the added influence of multiple (simulated or natural) turns. This is desirable because the Turn 3 response demonstrates the degree to which the SIMULATOR can continue a given conversation with the minimum amount of cues. Finally, this setup maximizes the evaluation data set from WildChat while still containing multiple turns, as most conversations in the dataset end with 3 or fewer turns.

4 Generating Simulation Data

To evaluate how well LLMs can be used to simulate human interactions, we generate a large dataset of dialogue simulations. We start with the WildChat dialogues between HUMANS and a CHATBOT (typically GPT-3.5) and then use another LLM as a SIMULATOR to continue the conversation.

Wildchat Data *WildChat* (Zhao et al., 2024) is a corpus of one million conversations between HUMANS and a CHATBOT, comprising over 2.5 million interaction turns in multiple languages, with English accounting for 53% of the turns. We restrict ourselves to a sample of 102k English instances, along with 10k Chinese and 10k Russian instances for multilingual experiments in §6.

To generate, we take the first two turns of each conversation (HUMAN initiation and CHATBOT response) as input to the SIMULATOR and prompt the model to generate what the HUMAN Turn 3 response would be, if any, or to indicate the human would not have responded.

Prompt Composition Since the wording of a prompt can have a significant impact on an LLM’s output and adopted persona (Röttger et al., 2024; Wright et al., 2024; Ceron et al., 2024), we conduct experiments with a wide range of prompts. Working independently, 12 prompt writers familiar with LLMs composed a total of 50 candidate prompts for this task. The prompt’s aim is to have the SIMULATOR match the conversational intent, content, and style of the first turn and to decide whether the conversation continues or not. Prompt writers were given 10 randomly sampled dialogues from Wild-

Chat to guide their efforts in prompt generation. They were encouraged to test their prompts using a CHATBOT or other available tools. Three example prompts are shown in Supplemental Table 3, and all 50 are in the Github repository.

Prompt Classification Prompts generally took one of three strategies: (1) **COT**: using chain-of-thought reasoning to guide the SIMULATOR through its response; (2) **OVERRIDE** employing various strategies to circumvent the SIMULATOR’s tendency to use overly polite, flowery, or long-winded language (e.g., jailbreaking, explicit instructions to ignore ethics training, jarring prompts); and (3) **DIRECT** directly instructing the SIMULATOR to respond like a human, including telling the SIMULATOR to pretend to be human or creating a persona for the model to follow (e.g., to-the-point, lazy). Two authors jointly labeled each prompt by strategy, resulting in 13 prompts as COT, 14 as OVERRIDE, and 23 as DIRECT.

Generating and Parsing Simulated Turn 3

Each of the 50 prompts was used to generate the third turn of a random sample of English-language dialogues from WildChat. Nine models were used: Meta-Llama-3.1-8B-Instruct (Dubey et al., 2024), Meta-Llama-3.1-70B-Instruct (Dubey et al., 2024), Meta-Llama-3-70B-Instruct (AI@Meta, 2024), Mistral-7B-Instruct-v0.3 (Jiang et al., 2023), Mixtral-8x7B-Instruct-v0.1 (Jiang et al., 2024a), Mistral-Large-Instruct (Mistral, 2024), Phi-3-medium-4k-instruct (Abdin et al., 2024), Qwen2-72B-Instruct (Yang et al., 2024), and c4ai-command-r-v01 (Cohere, 2024). We use customized regular expressions for each prompt to parse and extract the SIMULATOR’s response text, especially for the 13 COT prompts. After parsing, 4 prompts were discarded due to their frequent failure to produce valid output. Details on inference are in Appendix §B.

Evaluating LLM Simulations To evaluate, we select a breadth of 21 linguistic measures along 4 broad categories: style, lexical, syntactic, and semantic. These categories and measures have been broadly used for different NLP tasks (Sebastiani, 2002; Stamatos, 2009; Fu et al., 2018; Ribeiro et al., 2020). The output of these measures can be either a scalar, a probability distribution, or a feature vector. Scalars include both unbounded measures (e.g., utterance length) and bounded measures in $[0, 1]$ (e.g., text sentiment). Probability